

Question:

A 34-year-old poultry worker presents to his physician with a sore throat and a non-productive cough for 2 weeks. His cough is associated with fever. The vital signs include: blood pressure 120/80 mm Hg, heart rate 67/min, respiratory rate 18/min, and temperature 37.6°C (98.0°F). Physical examination shows oropharyngeal erythema and scattered, moist rales on lung auscultation. The patient’s X-ray demonstrates patchy reticular opacities in the perihilar regions of both lungs. After some additional tests, he is diagnosed with community-acquired pneumonia and is treated with cephalexin with modest improvement. Which of the following best describes the immune response elicited by the pathogen that is causing this patient’s condition?

Answer Choices:

- A. It activates TLR5 on the surface of macrophages.
- B. It primarily induces the Th1-cell response.
- C. Peptidoglycan is its major antigen that induces an immune response.
- D. This pathogen evades the immune response by encapsulation.

Reasoning + Answer:

< reasoning process here > The patient presents with symptoms of community -ac quired pneumonia (CAP), including sore throat , fever , non-productive cough , and physical exam findings of or oph ary nge al ery th ema and moist rales . Chest X -ray shows patch y ret icular op ac ities in the per ih ilar regions , which are classic for bacterial pneumonia . The patient is a poultry worker , which raises suspicion for exposure to common bacterial pathogens such as Stre pt oc occus pneumonia e , Ha em ophil us influenza e , or Mor ax ella cat arr halis . Among these , Stre pt oc occus pneumonia e is a leading cause of community -ac quired pneumonia , and it is a gram -positive bac illus with a cell wall containing pe pt id og ly can .

The immune response against such gram -positive bacteria is primarily mediated by Toll -like receptors (TLRs), particularly TLR 2 and TLR 6 , which recognize bacterial pe pt id og ly can components . TLR 5 is specific for flag ell in and is relevant to flag ell ated bacteria , such as certain enter ob acter i ace ae , not typical for gram -positive pathogens like S . pneumonia e . Therefore , option A (activ ates TLR 5 on macro ph ages) is incorrect .

P ne um oc occus (S . pneumonia e) primarily induces a Th 1 response , as it is a bacterial path ogen requiring op son ization , neutroph il recruitment , and cytokine production , such as interfer on -g amma , which is characteristic of Th 1 cells . The Th 1 response is critical in clearing intracellular and extracellular bacterial infections , including bacterial pneumonia . The path ogen triggers macro ph age and neutroph il activation , which is facilitated through cytokines associated with Th 1 immunity .

The pe pt id og ly can of S . pneumonia e is indeed a major antigen involved in triggering immune responses , such ÑhÑiÑÑH ly recognized via TLR s , but the question is not directly asking which molecule is the antigen . So while " pe pt id og ly can " is a correct antigen ic component , it does not best describe the *immune response* elic ited .

Regarding encaps ulation , encaps ulated organisms like S . pneumonia e do evade ph ag oc yt osis via their polys ac charide capsule , which inhib its op son ization . However , the question asks for the immune response *elic ited by* the path ogen , not how it ev ades immunity , making D incorrect as it shifts focus to immune evasion .

Therefore , the best description of the immune response is the activation of a Th 1 response , which is needed to clear intracellular or intracellular -like bacterial pathogens via cytotoxic and inflammatory responses .

Answer C , while fact ually correct about pe pt id og ly can being an antigen , is not the best description of the immune *response* , as it focuses on a component rather than the functional immune response .

Therefore , the most appropriate and accurate choice that describes the *immune response* is the Th 1 cell activation , as a hallmark in responding to bacterial pneumonia .

< answer > B . It primarily induces the Th 1 -cell response . </ answer >